

26 Rocket Propellants An experiment was conducted to compare three mixtures of components in making rocket propellant. Five samples of each mixture were obtained for testing. Five investigators measured the propellant thrust in pounds for each of the mixtures, with the mixtures presented in random order.

Investigators	Mixtures		
	1	2	3
1	42	39	45
2	37	36	40
3	43	48	50
4	41	47	52
5	39	42	47

- Identify the treatments and the blocks in this experiment and provide an appropriate analysis of variance.
- Is there sufficient evidence to conclude there is a significant difference in the propellant thrust for the mixtures? Use $\alpha = .01$.
- Was blocking effective? Use the p -value to draw your conclusions and justify your answer.
- Use Tukey's pairwise comparisons with $\alpha = .01$ to determine which propellant mixtures differed from others. Summarize your results.

ANOVA	SS	df	MS	F
SST	107.2	2	53.6	11.051
SSB	176.4	4	44.1	9.092
SSE	388	8	48.5	
SS	322.4	14		

$$c. P(F_{4,8} > 9.092) < 0.005 < 0.01$$

⇒ There is sufficient evidence to say there is a significant difference in means of different blocks.
block is effective

$$d. w = q_{0.01}(3,8) \sqrt{\frac{MSE}{b}}$$

$$= q_{0.01}(3,8) \sqrt{\frac{48.5}{5}}$$

$$= 5.5498$$

$$\begin{aligned} |T_1 - T_2| &= 2 < w \\ |T_2 - T_3| &= 4.4 < w \\ |T_1 - T_3| &= 6.4 > w \end{aligned} \Rightarrow \begin{aligned} &\text{mi and } m_2, m_2 \text{ and } m_3 \\ &\text{are not significantly different} \\ &m_1 \text{ and } m_3 \\ &\text{are significantly different.} \end{aligned}$$

a. Treatment = Mixtures

blocks : Investigators

$$b. T_1 = 202 \quad \sum x^2 = 8184 \quad \bar{T}_1 = 40.4$$

$$T_2 = 212 \quad \sum x^2 = 9094 \quad \bar{T}_2 = 42.4$$

$$T_3 = 234 \quad \sum x^2 = 11038 \quad \bar{T}_3 = 46.8$$

$$CM = \frac{G^2}{n} = \frac{(202 + 212 + 234)^2}{15} = 27993.6$$

$$SS = \sum x^2 - CM = 322.4$$

$$SST = \sum \frac{T_i^2}{n_i} - CM = 107.2$$

$$SSE = 322.4 - 107.2 = 215.2$$

$$B_1 = 126 \quad \sum x^2 = 5310 \quad B_4 = 140 \quad \sum x^2 = 6594$$

$$B_2 = 113 \quad \sum x^2 = 4265 \quad B_5 = 128 \quad \sum x^2 = 5494$$

$$B_3 = 141 \quad \sum x^2 = 6653$$

$$SSB = \sum \frac{B_i^2}{n_i} - CM = 176.4$$

$$H_0: \mu_1 = \mu_2 = \mu_3, H_a: \text{at least one difference}$$

$$\alpha = 0.01$$

$$F_{STAT} = \frac{MST}{MSE} \sim F_{2,8}$$

$$F^* = 11.051$$

$$RR = \{F | F > F_{2,8}\} = \{F | F > 8.65\}$$

$$11.051 > 8.65 \Rightarrow \text{Reject } H_0$$

⇒ There is a significant difference in the propellant thrust of the mixtures